



Series · Mark Scheme

Core Pure 1 Chapter 3: Series. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Standard summation formulae

- 1 (a) $\frac{1}{2}(20)(21) = 210$ B1; (b) $\frac{1}{6}(10)(11)(21) = 385$ B1.
 2 $\frac{1}{4}(12)^2(13)^2$ M1; = 6084 A1.
 3 $3\sum r^2 - 2\sum r$ M1; $3 \times 1240 - 2 \times 120$ M1; = 3480 A1.
 4 $6\sum r^2 + 2\sum r$ M1; $= n(n+1)(2n+1) + n(n+1)$ M1; $= n(n+1)(2n+2)$ M1; $= 2n(n+1)^2$ A1.
 5(a) $\sum r^2 + 2\sum r$ M1; $\frac{1}{6}n(n+1)(2n+1) + n(n+1)$ M1; $= \frac{1}{6}n(n+1)[(2n+1) + 6]$ M1; $= \frac{1}{6}n(n+1)(2n+7)$ A1.
 5(b) $n(n+1)(2n+7) = 24n^2 \Rightarrow (n+1)(2n+7) = 24n$ M1; $2n^2 - 15n + 7 = 0$, $(2n-1)(n-7) = 0$ M1; $n = 7$ (as n is a positive integer) A1.

Section B – Sums over a range

- 6 $\sum_{r=1}^{20} r - \sum_{r=1}^{10} r = 210 - 55$ M1; = 155 A1.
 7 $\sum_{r=1}^{20} r^3 - \sum_{r=1}^9 r^3$ M1; $44100 - 2025$ A1; = 42075 A1.
 8 $\frac{1}{6}(2n)(2n+1)(4n+1) - \frac{1}{6}n(n+1)(2n+1)$ M1; $= \frac{1}{6}n(2n+1)[2(4n+1) - (n+1)]$ M1; $= \frac{1}{6}n(2n+1)(7n+1)$ A1.
 9 $\sum (r^2 + 4r + 3)$ M1; $\frac{1}{6}n(n+1)(2n+1) + 2n(n+1) + 3n$ M1;
 $= \frac{1}{6}n[(n+1)(2n+1) + 12(n+1) + 18]$ M1; $= \frac{1}{6}n(2n^2 + 15n + 31)$ A1.
 10(a) $\sum (r^3 + 3r^2 + 2r)$ M1; $\frac{1}{4}n^2(n+1)^2 + \frac{1}{2}n(n+1)(2n+1) + n(n+1)$ M1; $= \frac{1}{4}n(n+1)[n(n+1) + 2(2n+1) + 4]$ M1;
 $= \frac{1}{4}n(n+1)(n^2 + 5n + 6) = \frac{1}{4}n(n+1)(n+2)(n+3)$ A1.
 10(b) $\frac{1}{4}(2n)(2n+1)(2n+2)(2n+3) - \frac{1}{4}n(n+1)(n+2)(n+3)$ M1; $= \frac{1}{4}n(n+1)[4(2n+1)(2n+3) - (n+2)(n+3)]$ M1;
 $= \frac{1}{4}n(n+1)(15n^2 + 27n + 6) = \frac{3}{4}n(n+1)(5n^2 + 9n + 2)$ A1.
 10(c) $n = 10$: $\frac{3}{4}(10)(11)(592)$ M1; = 48840 A1.

Section C – Method of differences

- 11 $\frac{A}{r} + \frac{B}{r+2}$, $2 = A(r+2) + Br$ M1; $\frac{1}{r} - \frac{1}{r+2}$ A1.
 12 $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots + (\frac{1}{n} - \frac{1}{n+1})$ M1; intermediate terms cancel to leave $1 - \frac{1}{n+1}$ M1; $= \frac{n}{n+1}$ A1.
 13 $r^2[(r+1)^2 - (r-1)^2] = r^2(4r)$ M1; $= 4r^3$ A1; $4\sum r^3 = f(n) - f(0) = n^2(n+1)^2$, so $\sum r^3 = \frac{1}{4}n^2(n+1)^2$ B1.
 14 $\frac{1}{(2r-1)(2r+1)} = \frac{1}{2} \left(\frac{1}{2r-1} - \frac{1}{2r+1} \right)$ M1 A1; $\frac{1}{2}[(1 - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{5}) + \dots] = \frac{1}{2}(1 - \frac{1}{2n+1})$ M1; $= \frac{n}{2n+1}$ A1.
 15(a) $1 = A(r+2) + Br$ M1; $\frac{1}{2r} - \frac{1}{2(r+2)}$ A1.
 15(b) $\frac{1}{2}[(1 - \frac{1}{3}) + (\frac{1}{2} - \frac{1}{4}) + (\frac{1}{3} - \frac{1}{5}) + \dots + (\frac{1}{n} - \frac{1}{n+2})]$ M1;
 $= \frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$ M1; $= \frac{3(n+1)(n+2) - 2(n+2) - 2(n+1)}{4(n+1)(n+2)}$ M1; $= \frac{3n^2 + 5n}{4(n+1)(n+2)} = \frac{n(3n+5)}{4(n+1)(n+2)}$ A1.
 15(c) $S_{20} - S_{10} = \frac{1300}{1848} - \frac{350}{528}$ M1; $= \frac{25}{616}$ A1.
 15(d) $\frac{3}{4}$ B1.



Roar-level challenge

16 $\frac{(r+2)-r}{r(r+1)(r+2)} M1; = \frac{2}{r(r+1)(r+2)} A1;$

sum $= \frac{1}{2} \left[\frac{1}{1 \times 2} - \frac{1}{(n+1)(n+2)} \right]$ by the method of differences $M1; = \frac{n(n+3)}{4(n+1)(n+2)} A1;$ sum to infinity $\frac{1}{4} B1.$